

IT131 PROJECT

Create

```sql
CREATE TABLE Patient (
patient_id INT PRIMARY KEY,
name VARCHAR(100) NOT NULL,
age INT,
gender VARCHAR(10),
phone VARCHAR(20),
address VARCHAR(255),
blood_type VARCHAR(5),
ward_id INT,
FOREIGN KEY (ward_id) REFERENCES Wardroom(ward_id) ;)
```

Patient table

blood_type	address	phone	gender	BirthDate	name	patient_id
A+	Riyadh, KSA	0551234567	Male	1980-01-01	Ahmed Ali	1
A+	Jeddah, KSA	0567654321	Female	1995-03-15	Sara Khaled	2
AB+	Riyadh, KSA	0578889990	Female	2000-09-10	Laila Mansour	4
A-	Mecca, KSA	0584445556	Male	1987-12-30	Faisal Saud	5
O-	Al Khobar, KSA	0591234567	Female	1990-07-12	Nada Saleh	6

Insert

```sql
INSERT INTO Patient (patient_id, name, age, gender, phone, address, blood_type, ward_id) VALUES
(1, 'Ahmed Ali', 45, 'Male', '0551234567', 'Riyadh, KSA', 'A+', 101),
(2, 'Sara Khaled', 30, 'Female', '0567654321', 'Jeddah, KSA', 'B-', 102),
(3, 'Omar Hassan', 60, 'Male', '0542223333', 'Dammam, KSA', 'O+', 101),
(4, 'Laila Mansour', 25, 'Female', '0578889990', 'Riyadh, KSA', 'AB+', 103),
(5, 'Faisal Saud', 38, 'Male', '0584445556', 'Mecca, KSA', 'A-', 102);
```

Create

```sql
CREATE TABLE Doctor (
doctor_id INT PRIMARY KEY,
name VARCHAR(100) NOT NULL,
specialization VARCHAR(100),
phone VARCHAR(20),
department_id INT,
FOREIGN KEY (department_id) REFERENCES Department(department_id) ;)
```

Doctor table

doctor_id	name	specialization	phone	department_id
1	Dr. Amal Nasser	Pediatrics	0522321111	201
2	Dr. Khaled Fahad	Cardiology	0553333444	202
3	Dr. Mousa Salem	Neurology	0556556666	203
4	Dr. Majed Omar	General Surgery	0549998888	204
5	Dr. Huda Ameen	Dermatology	0581239876	205

Insert

```sql
INSERT INTO Doctor (doctor_id, name, specialization, phone, department_id) VALUES
(1, 'Dr. Amal Nasser', 'Pediatrics', '0532221111', 201),
(2, 'Dr. Khaled Fahad', 'Cardiology', '0553334444', 202),
(3, 'Dr. Noura Salem', 'Neurology', '0565556666', 203),
(4, 'Dr. Majed Omar', 'General Surgery', '0549998888', 204),
(5, 'Dr. Huda Ameen', 'Dermatology', '0581239876', 205);
```

Create

```sql
CREATE TABLE Appointment (
appointment_id INT PRIMARY KEY,
patient_id INT,
doctor_id INT,
appointment_date DATE,
status VARCHAR(50),
FOREIGN KEY (patient_id) REFERENCES Patient(patient_id),
FOREIGN KEY (doctor_id) REFERENCES Doctor(doctor_id) ;)
```

Appointment table

status	appointment_date	doctor_id	patient_id	appointment_id
Active	2025-04-10	2	1	1
Completed	2025-04-11	1	2	2
Active	2025-04-13	4	4	4
Active	2025-04-14	5	5	5

Insert

```sql
INSERT INTO Appointment (appointment_id, patient_id, doctor_id, appointment_date, status) VALUES
(1, 1, 2, '2025-04-10', 'Active'),
(2, 2, 1, '2025-04-11', 'Completed'),
(3, 3, 3, '2025-04-12', 'Cancelled'),
(4, 4, 4, '2025-04-13', 'Active'),
(5, 5, 5, '2025-04-14', 'Active');
```

Create

```sql
CREATE TABLE Wardroom (
ward_id INT PRIMARY KEY,
ward_number VARCHAR(20),
capacity INT,
type VARCHAR(50) ;)
```

Wardroom TABLE

type	capacity	ward_number	ward_id
General	5	W-101	101
ICU	3	W-102	102
Pediatric	4	W-103	103
Maternity	6	W-104	104
Isolation	2	W-105	105

Insert

```sql
INSERT INTO Wardroom (ward_id, ward_number, capacity, type) VALUES
(101, 'W-101', 5, 'General'),
(102, 'W-102', 3, 'ICU'),
(103, 'W-103', 4, 'Pediatric'),
(104, 'W-104', 6, 'Maternity'),
(105, 'W-105', 2, 'Isolation');
```

Create

```sql
CREATE TABLE Nurse (
nurse_id INT PRIMARY KEY,
name VARCHAR(100),
shift VARCHAR(50),
ward_id INT,
FOREIGN KEY (ward_id) REFERENCES Wardroom(ward_id) ;)
```

Nurse table

ward_id	shift	name	nurse_id
101	Night	Amani Hassan	1
102	Morning	Salem Turki	2
103	Night	Reem Saeed	3
104	Evening	Nawal Majed	4
105	Morning	Mohammed Fahd	5

Insert

```sql
INSERT INTO Nurse (nurse_id, name, shift, ward_id) VALUES (1, 'Amani Hassan', 'Night', 101),
(2, 'Salem Turki', 'Morning', 102),
(3, 'Reem Saeed', 'Night', 103),
(4, 'Nawal Majed', 'Evening', 104),
(5, 'Mohammed Fahd', 'Morning', 105);
```

Create

```
CREATE TABLE Treatment (
treatment_id INT PRIMARY KEY,
patient_id INT,
doctor_id INT,
description TEXT,
treatment_date DATE,
FOREIGN KEY (patient_id) REFERENCES Patient(patient_id),
FOREIGN KEY (doctor_id) REFERENCES Doctor(doctor_id) ;)
```

Treatment table

treatment_date	description	doctor_id	patient_id	treatment_id
2025-04-09	Heart checkup	2	1	1
2025-04-08	Child vaccination	1	2	2
2025-04-06	Appendix surgery	4	4	4
2025-04-05	Skin allergy treatment	5	5	5

Insert

```
INSERT INTO Treatment (treatment_id, patient_id, doctor_id, description, treatment_date) VALUES
(1, 1, 2, 'Heart checkup', '2025-04-09'),
(2, 2, 1, 'Child vaccination', '2025-04-08'),
(3, 3, 3, 'MRI scan', '2025-04-07'),
(4, 4, 4, 'Appendix surgery', '2025-04-06'),
(5, 5, 5, 'Skin allergy treatment', '2025-04-05');
```

Create

```
CREATE TABLE Medicine (
medicine_id INT PRIMARY KEY,
name VARCHAR(100),
usage TEXT,
price DECIMAL(10,2)
;)
```

Medicine table

price	usage_description	name	medicine_id
11.00	Pain relief	Paracetamol	1
28.05	Antibiotic	Amoxicillin	2
49.50	Asthma inhaler	Ventolin	3
33.83	Allergy relief	Claritin	5

Insert

```
INSERT INTO Medicine (medicine_id, name, usage, price) VALUES
(1, 'Paracetamol', 'Pain relief', 10.0),
(2, 'Amoxicillin', 'Antibiotic', 25.5),
(3, 'Ventolin', 'Asthma inhaler', 45.0),
(4, 'Insulin', 'Diabetes treatment', 100.0),
(5, 'Claritin', 'Allergy relief', 30.75);
```

Create

```
CREATE TABLE Bill (
bill_id INT PRIMARY KEY,
patient_id INT,
treatment_id INT,
amount DECIMAL(10,2),
payment_status VARCHAR(50),
FOREIGN KEY (patient_id) REFERENCES Patient(patient_id),
FOREIGN KEY (treatment_id) REFERENCES Treatment(treatment_id) ;)
```

Bill table

payment_status	amount	treatment_id	patient_id	bill_id
Paid	500.00	1	1	1
Paid	250.00	2	2	2
Unpaid	1080.00	4	4	4
Paid	300.00	5	5	5
Unpaid	650.00	1	1	6

Insert

```
INSERT INTO Bill (bill_id, patient_id, treatment_id, amount, payment_status) VALUES
(1, 1, 1, 500.0, 'Paid'),
(2, 2, 2, 250.0, 'Unpaid'),
(3, 3, 3, 700.0, 'Paid'),
(4, 4, 4, 1200.0, 'Unpaid'),
(5, 5, 5, 300.0, 'Paid');
```

Create

```
CREATE TABLE Administrator (
admin_id INT PRIMARY KEY,
username VARCHAR(50) UNIQUE NOT NULL,
password VARCHAR(255) NOT NULL,
email VARCHAR(100) UNIQUE NOT NULL ;)
```

Administrator table

email	password	username	admin_id
admin1@hospital.com	pass123	admin1	1
manager2@hospital.com	123manager	manager2	2
sysadmin@hospital.com	secure456	sysadmin	3
control4@hospital.com	ctrl789	control4	4
root5@hospital.com	rootpass321	root5	5

Insert

```
INSERT INTO Administrator (admin_id, username, password, email) VALUES
(1, 'admin1', 'pass123', 'admin1@hospital.com'),
(2, 'manager2', '123manager', 'manager2@hospital.com'),
(3, 'sysadmin', 'secure456', 'sysadmin@hospital.com'),
(4, 'control4', 'ctrl789', 'control4@hospital.com'),
(5, 'root5', 'rootpass321', 'root5@hospital.com');
```

Create

```
CREATE TABLE Department ( department_id INT PRIMARY KEY,
name VARCHAR(100),
location VARCHAR(100)
;)
```

Department table

location	name	department_id
Floor 1	Pediatrics	201
Floor 2	Cardiology	202
Floor 3	Neurology	203
Floor 4	Surgery	204
Floor 5	Dermatology	205

Insert

```
INSERT INTO Department (department_id, name, location) VALUES
(201, 'Pediatrics', 'Floor 1'),
(202, 'Cardiology', 'Floor 2'),
(203, 'Neurology', 'Floor 3'),
(204, 'Surgery', 'Floor 4'),
(205, 'Dermatology', 'Floor 5');
```

Create View (ActiveAppointments)

```sql
CREATE VIEW ActiveAppointments AS
SELECT * FROM Appointment
WHERE status = 'Active'
WITH CHECK OPTION;
```

appointment_id	patient_id	doctor_id	appointment_date	status
1	1	2	2025-04-10	Active
3	3	4	2025-04-13	Active
4	4	3	2025-04-14	Active

Create View (AvailableWards)

```
CREATE VIEW AvailableWards AS
SELECT * FROM Wardroom WHERE capacity > 0
WITH CHECK OPTION;
```

ward_id	ward_number	capacity	type
101	W-101	5	General
102	W-102	3	ICU
103	W-103	4	Pediatric
104	W-104	6	Maternity
105	W-105	2	Isolation

Create View (NightShiftNurses)

```sql
CREATE VIEW NightShiftNurses AS
SELECT * FROM Nurse
WHERE shift = 'Night'
WITH CHECK OPTION;
```

nurse_id	name	shift	ward_id
1	Amani Hassan	Night	101
3	Reem Saeed	Night	103

Create View (UnpaidBills)

```sql
CREATE VIEW UnpaidBills AS
SELECT * FROM Bill
WHERE payment_status = 'Unpaid'
WITH CHECK OPTION;
```

bill_id	patient_id	treatment_id	amount	payment_status
2	2	2	250.0	Unpaid
4	4	4	1200.0	Unpaid

Create View (PediatricDoctors)

```sql
CREATE VIEW PediatricDoctors AS
SELECT * FROM Doctor
WHERE specialization = 'Pediatrics'
WITH CHECK OPTION;
```

doctor_id	name	specialization	phone	department_id
1	Dr. Amal Nasser	Pediatrics	0522321111	201

Queries

1

```
SELECT name FROM Doctor WHERE department_id = 202
```

NAME
Dr. Khaled Fahad

2

```
SELECT name, gender, BirthDate FROM Patient
```

NAME	GENDER	BIRTHDATE
Ahmed Ali	Male	1980-01-01
Sara Khaled	Female	1995-03-15
Omar Hassan	Male	1965-06-25
Laila Mansour	Female	2000-09-10
Faisal Saud	Male	1987-12-30

3

```
SELECT name, price FROM Medicine WHERE price BETWEEN 30 AND 100
```

NAME	PRICE
Ventolin	49.5
Claritin	33.83

Queries

4

```
SELECT AVG(price) AS average_price FROM Medicine
```

AVERAGE_PRICE
30.595

5

```
SELECT amount FROM Bill ORDER BY amount DESC
```

AMOUNT
1200
700
500
300
250

6

```
SELECT username, email FROM Administrator WHERE email LIKE '%.com'
```

USERNAME	EMAIL
admin1	admin1@hospital.com
manager2	manager2@hospital.com
sysadmin	sysadmin@hospital.com
control4	control4@hospital.com
root5	root5@hospital.com

Queries

7

```
SELECT username, email FROM Administrator WHERE email LIKE '%@hospital.com'
```

USERNAME	EMAIL
admin1	admin1@hospital.com
manager2	manager2@hospital.com
sysadmin	sysadmin@hospital.com
control4	control4@hospital.com
root5	root5@hospital.com

8

```
SELECT name, specialization, Salary FROM Doctor ORDER BY Salary DESC
```

NAME	SPECIALIZATION	SALARY
Dr. Noura Salem	Neurology	95000
Dr. Khaled Fahad	Cardiology	90000
Dr. Huda Ameen	Dermatology	78000
Dr. Tarek Samir	Orthopedics	60000
Dr. Amal Nasser	Pediatrics	45000
Dr. Majed Omar	Surgery	12000

9

```
SELECT name, price FROM Medicine WHERE price BETWEEN 20 AND
```

NAME	PRICE
Amoxicillin	25.5
Ventolin	45
Claritin	30.75

Queries

10

```
SELECT name, blood_type FROM Patient WHERE blood_type = 'A+'
```

NAME	BLOOD_TYPE
Ahmed Ali	A+
Sara Khaled	A+

11

```
SELECT name, blood_type, BirthDate FROM Patient
```

NAME	BLOOD_TYPE	BIRTHDATE
Ahmed Ali	A+	1980-01-01
Sara Khaled	A+	1995-03-15
Laila Mansour	AB+	2000-09-10
Faisal Saud	A-	1987-12-30
Nada Saleh	O-	1990-07-12

12

```
SELECT name, Salary FROM Doctor WHERE department_id = 204
```

NAME	SALARY
Dr. Majed Omar	12000
Dr. Tarek Samir	60000

13

```
SELECT p.name AS patient_name, SUM(b.amount) AS total_bill FROM Patient p JOIN Bill b ON p.patient_id = b.patient_id GROUP BY p.name
```

PATIENT_NAME	TOTAL_BILL
Laila Mansour	1080
Sara Khaled	250
Ahmed Ali	1150
Faisal Saud	300

Queries

15

```
SELECT name, Salary FROM Doctor WHERE Salary > 80000
```

NAME	SALARY
Dr. Khaled Fahad	90000
Dr. Noura Salem	95000

16

```
SELECT n.name, n.shift FROM Nurse n JOIN Wardroom w ON n.ward_id = w.ward_id WHERE w.ward_number = 'W-103'
```

NAME	SHIFT
Reem Saeed	Night

14

```
SELECT name, specialization FROM Doctor WHERE department_id = 205
```

NAME	SPECIALIZATION
Dr. Huda Ameen	Dermatology

17

```
SELECT name, CASE WHEN shift = 'Night' THEN 'Night Shift' WHEN shift = 'Morning' THEN 'Morning Shift' ELSE 'Other Shift' END AS Shift_Type
```

NAME	SHIFT_TYPE
Amani Hassan	Night Shift
Salem Turki	Morning Shift
Reem Saeed	Night Shift
Nawal Majed	Other Shift
Mohammed Fahd	Morning Shift

Queries

18

```
SELECT name, phone FROM Patient WHERE patient_id IN (SELECT patient_id FROM Bill WHERE payment_status = 'Unpaid')
```

NAME	PHONE
Sara Khaled	0567654321
Laila Mansour	0578889990

19

```
SELECT p.name AS patient_name, d.name AS doctor_name, a.appointment_date FROM Appointment a JOIN Patient p ON a.patient_id = p.patient_id
```

PATIENT_NAME	DOCTOR_NAME	APPOINTMENT_DATE
Sara Khaled	Dr. Amal Nasser	11-APR-25
Ahmed Ali	Dr. Khaled Fahad	10-APR-25
Laila Mansour	Dr. Majed Omar	13-APR-25
Faisal Saud	Dr. Huda Ameen	14-APR-25

20

```
UPDATE Medicine
;SET price = price * 1.1
;select* from Medicine
```

medicine_id	name	usage_description	price
1	Paracetamol	Pain relief	11.00
2	Amoxicillin	Antibiotic	28.05
3	Ventolin	Asthma inhaler	49.50
5	Claritin	Allergy relief	33.83

Queries

21

```
SELECT name, email
FROM Administrator
WHERE email LIKE '%admin%';
```

name	email
admin1	admin1@hospital.com
sysadmin	sysadmin@hospital.com

22

```
SELECT d.name AS doctor_name,
COUNT(a.appointment_id) AS total_appointments
FROM Doctor d
LEFT JOIN Appointment a ON d.doctor_id = a.doctor_id
GROUP BY d.name;
```

doctor_name	total_appointments
Dr. Amal Nasser	1
Dr. Huda Ameen	1
Dr. Khaled Fahad	1
Dr. Majed Omar	1
Dr. Noura Salem	0
Dr. Tarek Samir	0

23

```
SELECT name, amount
FROM Bill
ORDER BY amount ASC;
```

name	amount
Sara Khaled	250.00
Faisal Saud	300.00
Ahmed Ali	500.00
Ahmed Ali	650.00
Laila Mansour	1080.00

24

```
UPDATE Patient
SET blood_type = 'A+'
WHERE patient_id = 2;
```

name	blood_type
Sara Khaled	A+

Queries

25

SELECT name, usage
FROM Medicine
WHERE price < 50;

name	usage
Paracetamol	Pain relief
Amoxicillin	Antibiotic
Ventolin	Asthma inhaler
Claritin	Allergy relief

26

SELECT name, phone, BirthDate
FROM Patient;

name	phone	BirthDate
Ahmed Ali	0551234567	1980-01-01
Sara Khaled	0567654321	1995-03-15
Laila Mansour	0578889990	2000-09-10
Faisal Saud	0584445556	1987-12-30
Nada Saleh	0591234567	1990-07-12

27

SELECT name, usage
FROM Medicine
WHERE price BETWEEN 20 AND 60;

name	usage
Amoxicillin	Antibiotic
Ventolin	Asthma inhaler
Claritin	Allergy relief

28

SELECT * FROM ActiveAppointments;

appointment_id	patient_id	doctor_id	appointment_date	status
1	1	2	2025-04-10	Active
4	4	4	2025-04-13	Active
5	5	5	2025-04-14	Active

Queries

29

```
DELETE FROM Appointment
WHERE status = 'Cancelled';
```

appointment_id	patient_id	doctor_id	appointment_date	status
1	1	2	2025-04-10	Active
2	2	1	2025-04-11	Completed
4	4	4	2025-04-13	Active
5	5	5	2025-04-14	Active

30

```
DELETE FROM Medicine
WHERE price > 90;
```

medicine_id	name	usage_description	price
1	Paracetamol	Pain relief	11.00
2	Amoxicillin	Antibiotic	28.05
3	Ventolin	Asthma inhaler	49.50
5	Claritin	Allergy relief	33.83

Queries

31

```
DELETE FROM Bill
WHERE patient_id = 3;
DELETE FROM Bill
WHERE bill_id = 2;
```

bill_id	patient_id	treatment_id	amount	payment_status
1	1	1	500.00	Paid
4	4	4	1080.00	Unpaid
5	5	5	300.00	Paid
6	1	1	617.50	Unpaid

32

```
DELETE FROM Treatment
WHERE patient_id = 3;
```

treatment_id	patient_id	doctor_id	description	treatment_date
1	1	2	Heart checkup	2025-04-09
2	2	1	Child vaccination	2025-04-08
4	4	4	Appendix surgery	2025-04-06
5	5	5	Skin allergy treatment	2025-04-05

33

```
SELECT * FROM Patient WHERE gender = 'Male';
```

patient_id	name	BirthDate	gender	phone	address	blood_type
1	Ahmed Ali	1980-01-01	Male	0551234567	Riyadh	A+

Queries

34

SELECT name, specialization FROM Doctor WHERE Salary > 40000;

name	specialization
Dr. Amal Nasser	Pediatrics

35

SELECT COUNT(*) AS total_appointments FROM Appointment;

total_appointments
1

36

SELECT name, specialization FROM Doctor WHERE Salary > 40000;

name	specialization
Dr. Amal Nasser	Pediatrics
Dr. Khaled Fahad	Cardiology
Dr. Noura Salem	Neurology
Dr. Huda Ameen	Dermatology

37

SELECT * FROM Nurse WHERE shift = 'Night';

nurse_id	name	shift	ward_id
1	Amani Hassan	Night	101
3	Reem Saeed	Night	103

Queries

38

```
SELECT name FROM Medicine WHERE price < 20;
```

name
Paracetamol

39

```
SELECT name FROM Department
WHERE location = 'Floor 1';
```

name
Pediatrics

40

```
SELECT
COUNT(*)
FROM Patient;
```

COUNT(*)
5

41

```
SELECT * FROM Treatment
WHERE treatment_date = TO_DATE('2025-04-09', 'YYYY-MM-DD');
```

treatment_id	patient_id	doctor_id	description	treatment_date
1	1	2	Heart checkup	2025-04-09

Queries

42

SELECT name FROM Doctor WHERE DoctorGender = 'Female';

name
Dr. Amal Nasser
Dr. Noura Salem
Dr. Huda Ameen

43

SELECT patient_id FROM Bill GROUP BY patient_id HAVING COUNT(*) > 0;

patient_id
1
2
3
4
5

44

SELECT * FROM Wardroom WHERE capacity >= 5;

ward_id	ward_number	capacity	type
101	W-101	5	General
104	W-104	6	Maternity

45

SELECT name, price FROM Medicine WHERE name LIKE 'I%';

name	price
Insulin	110.00

Queries

46

```
SELECT name
FROM Patient
WHERE name LIKE '%Ali%';
```

name
Ahmed Ali

47

```
SELECT name, usage
FROM Medicine
WHERE price BETWEEN 15 AND 45;
```

name	usage
Amoxicillin	Antibiotic
Ventolin	Asthma inhaler
Claritin	Allergy relief

48

```
SELECT BillID, PatientID, TotalAmount, Status
FROM Bill
;'WHERE Status = 'Unpaid
```

Status	TotalAmount	PatientID	BillID
Unpaid	850.00	2001	301
Unpaid	430.00	2006	305

Queries

49

```
DELETE FROM Nurse
WHERE Shift = 'Night' AND Experience
;2 >
SELECT NurseID, Name, Shift, Experience FROM Nurse;
```

Experience	Shift	Name	NurseID
3	Day	Sarah K.	N101
5	Night	Omar H.	N103

50

```
SELECT DeptID, DeptName, Head
FROM Department
;'%WHERE DeptName LIKE '%Cardio
```

Head	DeptName	DeptID
Dr. Ali Z.	Cardiology	D01
Dr. Nada R.	Pediatric Cardio	D07

Normalisation Theory

Patient table

- 0NF
- Patient table

patient_id	name	age	gender	phone	address	blood_type	ward_id
1	Ahmed Ali	45	Male	0551234567	Riyadh, KSA	A+	101
2	Sara Khaled	30	Female	0567654321	Jeddah, KSA	B-	102
3	Omar Hassan	60	Male	0542223333	Dammam, KSA	O+	101
4	Laila Mansour	25	Female	0578889990	Riyadh, KSA	AB+	103
5	Faisal Saud	38	Male	0584445556	Mecca, KSA	A-	102

- 1NF
- The condition for 1NF is that each cell must contain a single atomic value. there are no columns with multiple values, and each cell contains a single, indivisible value. Therefore, thetable satisfies 1NF and does not require any changes

Normalisation Theory

- 2NF
- Patient table

patient_id	name	age	gender	phone	address	blood_type	ward_id
1	Ahmed Ali	45	Male	0551234567	Riyadh, KSA	A+	101
2	Sara Khaled	30	Female	0567654321	Jeddah, KSA	B-	102
3	Omar Hassan	60	Male	0542223333	Dammam, KSA	O+	101
4	Laila Mansour	25	Female	0578889990	Riyadh, KSA	AB+	103
5	Faisal Saud	38	Male	0584445556	Mecca, KSA	A-	102

- The Patient table is in 2NF because all columns depend entirely on the primary key (patientid), with no partial dependencies.

Normalisation Theory

- 3NF
- Patient table

patient_id	name	age	gender	phone	address	blood_type	ward_id
1	Ahmed Ali	45	Male	0551234567	Riyadh, KSA	A+	101
2	Sara Khaled	30	Female	0567654321	Jeddah, KSA	B-	102
3	Omar Hassan	60	Male	0542223333	Dammam, KSA	O+	101
4	Laila Mansour	25	Female	0578889990	Riyadh, KSA	AB+	103
5	Faisal Saud	38	Male	0584445556	Mecca, KSA	A-	102

- PatientContact

patient_id	phone	address
1	0551234567	Riyadh, KSA
2	0567654321	Jeddah, KSA
3	0542223333	Dammam, KSA
4	0578889990	Riyadh, KSA
5	0584445556	Mecca, KSA

Normalisation Theory

Treatment table

- 0NF
- Treatment table

treatment_id	patient_id	doctor_id	description	treatment_date
1	1	2	Heart checkup	2025-04-09
2	2	1	Child vaccination	2025-04-08
3	3	3	MRI scan	2025-04-07
4	4	4	Appendix surgery	2025-04-06
5	5	5	Skin allergy treatment	2025-04-05

- 1NF
- The table satisfies 1NF because each cell contains a single atomic value, and there are no columns with multiple values

Normalisation Theory

- 2NF
- Treatment table

description_id	description
1	Heart checkup
2	Child vaccination
3	MRI scan
4	Appendix surgery
5	Skin allergy treatment

- TreatmentDescription

treatment_id	patient_id	doctor_id	description_id	treatment_date
1	1	2	1	2025-04-09
2	2	1	2	2025-04-08
3	3	3	3	2025-04-07
4	4	4	4	2025-04-06
5	5	5	5	2025-04-05

- 3NF
- The Treatment table is in 3NF because all columns depend only
- on the primary key, and there are no transitive dependencies between non-key columns.

Normalisation Theory

Doctor table

- 0NF
- Doctor table

doctor_id	name	specialization	phone	department_id
1	Dr. Amal Nasser	Pediatrics	0532221111	201
2	Dr. Khaled Fahad	Cardiology	0553334444	202
3	Dr. Noura Salem	Neurology	0565556666	203
4	Dr. Majed Omar	General Surgery	0549998888	204
5	Dr. Huda Ameen	Dermatology	0581239876	205

- 1NF
- The Doctor table satisfies 1NF because each cell containsa single atomic value, and there areno columns withmultiple values.

Normalisation Theory

- 2NF
- Doctor table

department_id	specialization_name
201	Pediatrics
202	Cardiology
203	Neurology
204	General Surgery
205	Dermatology

- Specialization

doctor_id	name	phone	department_id
1	Dr. Amal Nasser	0532221111	201
2	Dr. Khaled Fahad	0553334444	202
3	Dr. Noura Salem	0565556666	203
4	Dr. Majed Omar	0549998888	204
5	Dr. Huda Ameen	0581239876	205

Normalisation Theory

- 3NF
- Doctor phone

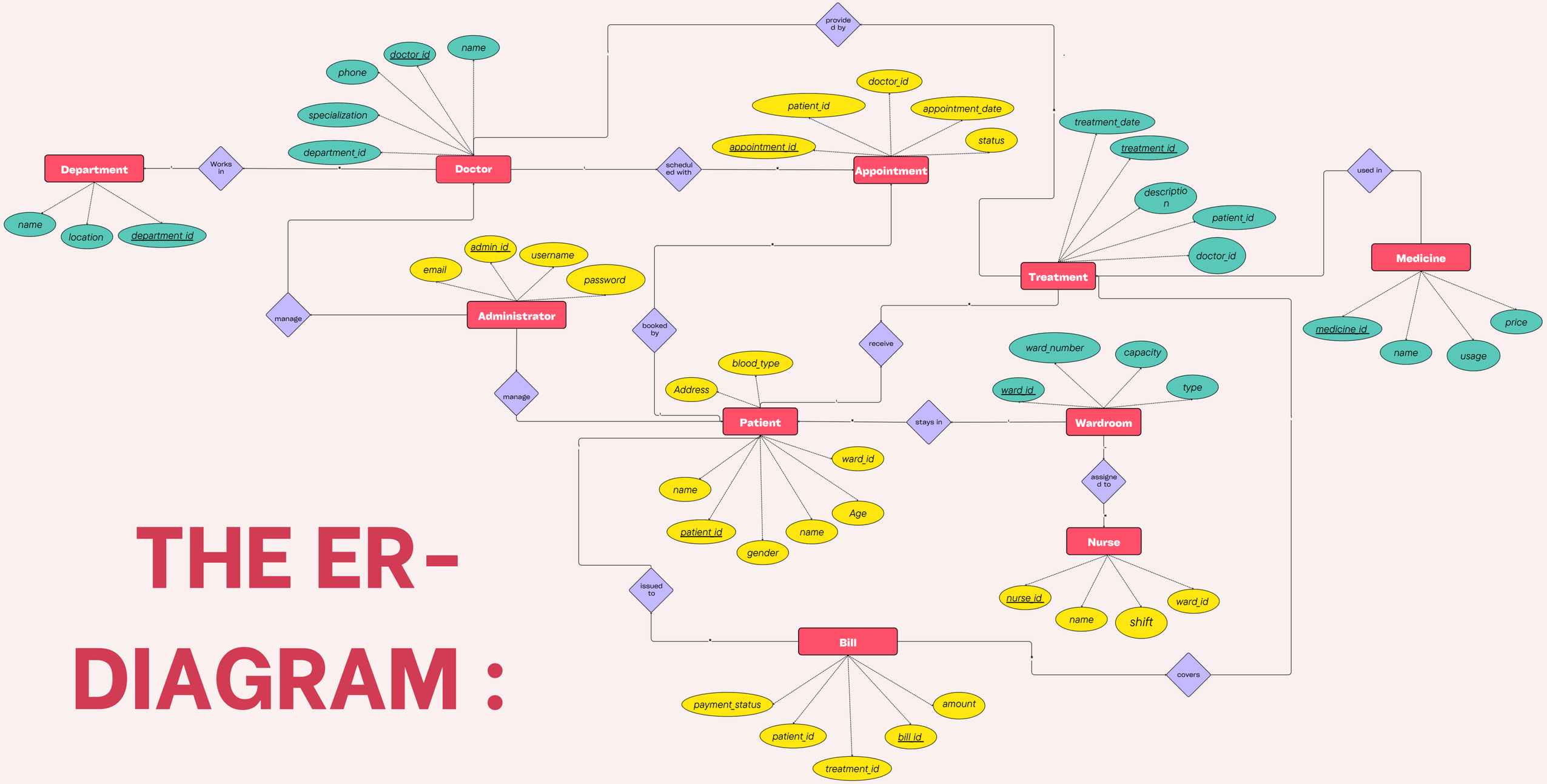
phone_id	doctor_id	phone
1	1	0532221111
2	2	0553334444
3	3	0565556666
4	4	0549998888
5	5	0581239876

- Specialization

specialization_id	specialization_name
1	Pediatrics
2	Cardiology
3	Neurology
4	General Surgery
5	Dermatology

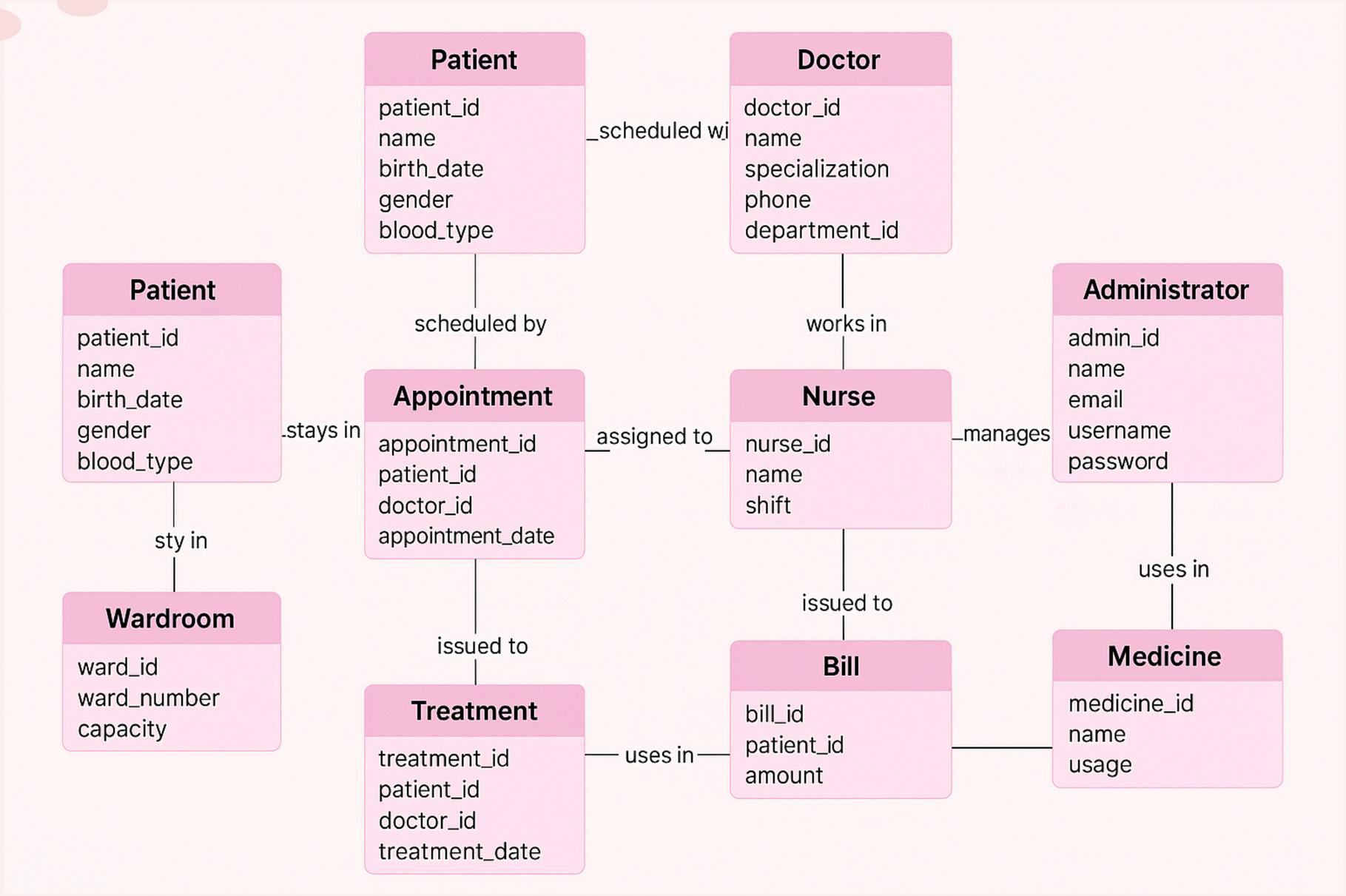
- Doctor

doctor_id	name	specialization_id
1	Dr. Amal Nasser	1
2	Dr. Khaled Fahad	2
3	Dr. Noura Salem	3
4	Dr. Majed Omar	4
5	Dr. Huda Ameen	5



THE ER-DIAGRAM :

The schema


Patient
patient_id
name
birth_date
gender
blood_type
_scheduled wi
Doctor
doctor_id
name
specialization
phone
department_id
Patient
patient_id
name
birth_date
gender
blood_type
scheduled by
works in
Administrator
admin_id
name
email
username
password
stays in
Appointment
appointment_id
patient_id
doctor_id
appointment_date
assigned to
Nurse
nurse_id
name
shift
manages
sty in
uses in
Wardroom
ward_id
ward_number
capacity
issued to
issued to
Bill
bill_id
patient_id
amount
Medicine
medicine_id
name
usage
Treatment
treatment_id
patient_id
doctor_id
treatment_date
uses in

Student Names and IDs

Name: رولا الغانم – ID: 451203244
Name: مضاوي السعيد – ID: 443200160
Name: جنان الشايع – ID: 451203368
Name: رنا – ID: 443213304
Name: أنفال السويد – ID: 442201248
Name: رزان الشتوي – ID: 441203662
Name: نشمية المطيري – ID: 442201389
Name: أريام المطيري – ID: 443213056

